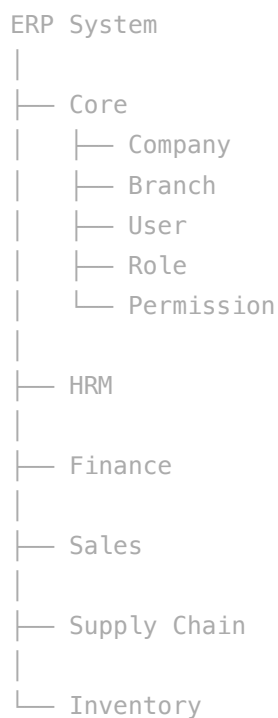


suppose you are a software eng
now i want to make an erp software
(python,django,django orm,html,tailwind css,postgres
sql,validation,mvt architecture,multiapp)
there is 4 module
Finance,sales,supply and chain, HRM, inventory
its an proffessional software development proccess
so at first i want to build the db design
in a veru proffessional way give me the db design

যদি তুমি একটি **professional ERP (Enterprise Resource Planning)** software বানাতে চাও,
তাহলে প্রথমে module-based database design করতে হবে।

আমি suggest করবো:

Core Architecture



Phase 1: Core Tables

সব module এগুলোর উপর depend করবে।

Company

SQL

Company

id (PK)

name

trade_license

tin

email

phone

address

website

status

created_at

updated_at

Branch

SQL

Branch

id (PK)

company_id (FK)

name

code

address

phone

email

status

User

Django User extend করবে।

SQL

User

id (PK)

employee_id (FK nullable)

username

email

```
password
is_active
is_staff
created_at
```

Role

SQL

```
Role
-----
id
name
description
```

Examples:

- Admin
- Finance Manager
- Sales Manager
- Store Manager
- HR Manager

Permission

SQL

```
Permission
-----
id
code
name
module
```

Example:

```
sales_create
sales_update

finance_view
finance_edit
```

UserRole

Many To Many

SQL

```
UserRole
-----
id
user_id
role_id
```

HRM Module

Department

SQL

```
Department
-----
id
name
description
```

Designation

SQL

```
Designation
-----
id
name
department_id
```

Employee

SQL

```
Employee
-----
```

id
employee_code
first_name
last_name
email
phone
gender
dob
joining_date
designation_id
department_id
branch_id
salary
status

Attendance

SQL

Attendance

id
employee_id
date
check_in
check_out
working_hours
status

Status:

Present
Absent
Leave
Late

LeaveType

SQL

LeaveType

id

name
days_allowed

LeaveRequest

SQL

LeaveRequest

id
employee_id
leave_type_id
start_date
end_date
reason
status
approved_by

Payroll

SQL

Payroll

id
employee_id
month
basic_salary
allowance
bonus
deduction
net_salary
paid_status

Inventory Module

Inventory হলো ERP এর heart.

Category

SQL

```
Category
-----
id
name
parent_id
```

Self Reference.

Unit

SQL

```
Unit
-----
id
name
symbol
```

Example:

```
Piece
KG
Liter
Box
```

Product

SQL

```
Product
-----
id
sku
barcode
name
description
category_id
unit_id
cost_price
selling_price
reorder_level
status
```

Warehouse

SQL

```
Warehouse
-----
id
branch_id
name
location
```

Stock

SQL

```
Stock
-----
id
product_id
warehouse_id
quantity
reserved_qty
available_qty
```

StockMovement

Inventory audit trail

SQL

```
StockMovement
-----
id
product_id
warehouse_id
movement_type
quantity
reference_type
reference_id
movement_date
```

Movement Types:

IN
OUT
TRANSFER
ADJUSTMENT

Supply Chain Module

Supplier

SQL

Supplier

id
name
contact_person
phone
email
address

PurchaseOrder

SQL

PurchaseOrder

id
po_number
supplier_id
order_date
expected_date
status
total_amount

PurchaseOrderItem

SQL

```
PurchaseOrderItem
-----
id
purchase_order_id
product_id
quantity
unit_price
subtotal
```

GoodsReceipt

```
SQL

GoodsReceipt
-----
id
purchase_order_id
warehouse_id
received_date
received_by
```

GoodsReceiptItem

```
SQL

GoodsReceiptItem
-----
id
goods_receipt_id
product_id
received_qty
```

Sales Module

Customer

```
SQL

Customer
-----
```

```
id
customer_code
name
phone
email
address
credit_limit
```

SalesOrder

SQL

```
SalesOrder
```

```
-----
```

```
id
order_number
customer_id
order_date
status
total_amount
```

Status:

```
Draft
Confirmed
Delivered
Cancelled
```

SalesOrderItem

SQL

```
SalesOrderItem
```

```
-----
```

```
id
sales_order_id
product_id
quantity
unit_price
discount
subtotal
```

Delivery

SQL

```
Delivery
-----
id
sales_order_id
delivery_date
status
```

Invoice

SQL

```
Invoice
-----
id
invoice_number
customer_id
sales_order_id
invoice_date
subtotal
tax
discount
grand_total
status
```

Payment

SQL

```
Payment
-----
id
invoice_id
customer_id
payment_method
amount
payment_date
reference_no
```

Finance Module

Accounting ERP-এর সবচেয়ে critical অংশ।

ChartOfAccounts

SQL

```
ChartOfAccounts
-----
id
account_code
account_name
account_type
parent_account
```

Account Types:

- Asset
- Liability
- Equity
- Revenue
- Expense

JournalEntry

SQL

```
JournalEntry
-----
id
voucher_no
entry_date
description
created_by
```

JournalEntryLine

Double Entry Accounting

SQL

```
JournalEntryLine
-----
id
journal_entry_id
account_id
debit
credit
```

FiscalYear

```
SQL

FiscalYear
-----
id
year_name
start_date
end_date
status
```

Expense

```
SQL

Expense
-----
id
expense_type
account_id
amount
expense_date
description
```

Revenue

```
SQL

Revenue
-----
id
account_id
amount
source
```

date
description

Professional ERP Relationships

Company

└─ Branch

Branch

└─ Employee

└─ Warehouse

└─ Sales

Department

└─ Employee

Designation

└─ Employee

Supplier

└─ PurchaseOrder

└─ PurchaseOrderItem

PurchaseOrder

└─ GoodsReceipt

└─ StockMovement(IN)

Product

└─ PurchaseOrderItem

└─ SalesOrderItem

└─ Stock

└─ StockMovement

Customer

└─ SalesOrder

└─ Invoice

SalesOrder

└─ SalesOrderItem

└─ Delivery

└─ Invoice

Invoice

└─ Payment

JournalEntry

└─ JournalEntryLine

এটা **ERP V1 database design**। Professional software company হলে আমি পরের ধাপে এটাকে আরও normalize করে:

- Audit Log
- Approval Workflow
- Notification System
- Multi Company
- Multi Currency
- Tax Engine (VAT/TAX)
- Cost Center
- Budget Management
- Inventory Batch/Lot Tracking
- Serial Number Tracking
- General Ledger (GL)
- Accounts Receivable (AR)
- Accounts Payable (AP)

যোগ করতাম, যেটা ERPNext/Odoo/SAP-এর architecture-এর কাছাকাছি হবে।

আমার মতে coding শুরু করার আগে **ER Diagram + PostgreSQL schema + Django model architecture (multi-app MVT)** finalize করা উচিত। এই design-এর ওপর সরাসরি Django ORM models তৈরি করা যাবে।



